

Radiation & Energy Balance

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Fundamental Concepts of Energy and Temperature



“Energy tells us how much work can be done,
while power tells us how fast that energy is used”

1. **Energy** is the capacity of a system to perform work or transfer heat. SI Unit: **Joule (J)**
2. **Power** is the rate of energy transfer or energy conversion. SI Unit: **Watt (W)**

where,

$$P = \frac{dE}{dt} \quad \text{eq. 1}$$

P = Power (W)
 E = Energy (J)
 t = Time (s)

3. Internal Energy and Temperature:

- Internal energy arises from the random motion of atoms and molecules.
- Temperature quantifies the average kinetic energy of these particles.
- Higher molecular motion corresponds to higher temperature.
- Internal Energy (U) $\propto$ Mean Kinetic Energy of constituent atoms/molecules.

$$T \text{ (K)} = T \text{ (}^\circ\text{C)} + 273.15 \quad \text{eq. 2}$$

Case Study:



Objective: Estimate the steady-state power output of a human body from daily caloric intake.

Given:

Average daily intake: $\dot{E} = 2000$ food calories/day

Conversion factor: 1 Cal = 4184 J

Assumption: energy intake $\approx$ energy expended (steady-state, no net mass change)

Step 1: Convert Energy to Joules:

$$E = 2000 \times 4184 = 8.368 \times 10^6 \text{ J/day}$$

Step 2: Convert Time base to seconds:

$$t = 24 \times 3600 = 8.64 \times 10^4 \text{ s}$$

Case Study:



Step 3: Average Power:

$$P = \frac{E}{t} = \frac{8.368 \times 10^6}{8.64 \times 10^4} \approx 97 \text{ W}$$

Interpretation:

- $P \approx 100\text{W}$, comparable to a standard incandescent light bulb
- Equivalent to $\approx 1/7^{\text{th}}$ horsepower (1 hp = 746 W)
- Equivalent to powering a 100-W incandescent light bulb



Electromagnetic Radiation

- Electromagnetic (EM) radiation is energy propagated through oscillating electric and magnetic fields.
- Nearly all energy received by the Earth from the Sun arrives as electromagnetic radiation.
- Unlike conduction and convection, EM radiation can travel through a vacuum, enabling energy transfer from the Sun to Earth.

➤ **Energy intensity (Power):** Energy per unit time
SI Unit: W (J/s)

➤ **Energy flux:** Intensity per unit area
SI Unit: W/m^2

Wave - Particle Duality of EM Radiation:

1. **Particle nature:** radiation as a stream of photons (discrete energy packets)
Emitted when an electron transitions from a higher to a lower energy state
2. **Wave nature:** characterized by wavelength (λ) and frequency (ν)

Electromagnetic Spectrum

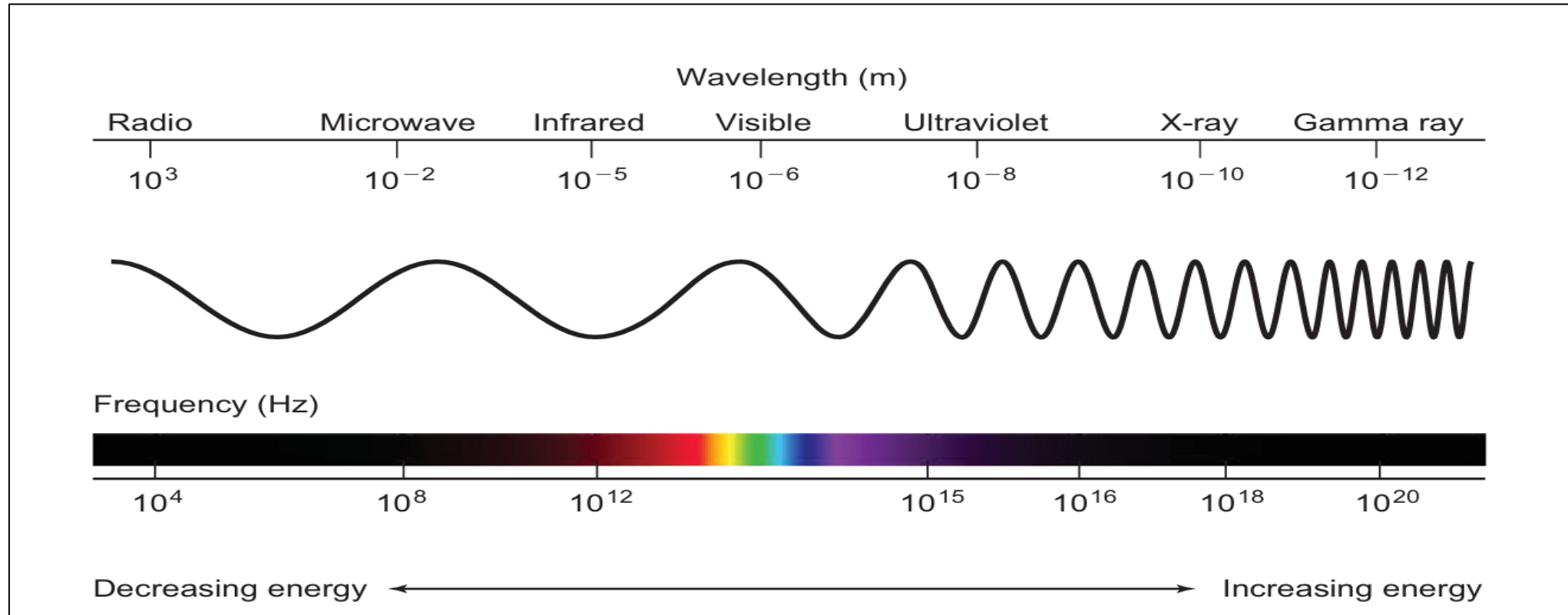
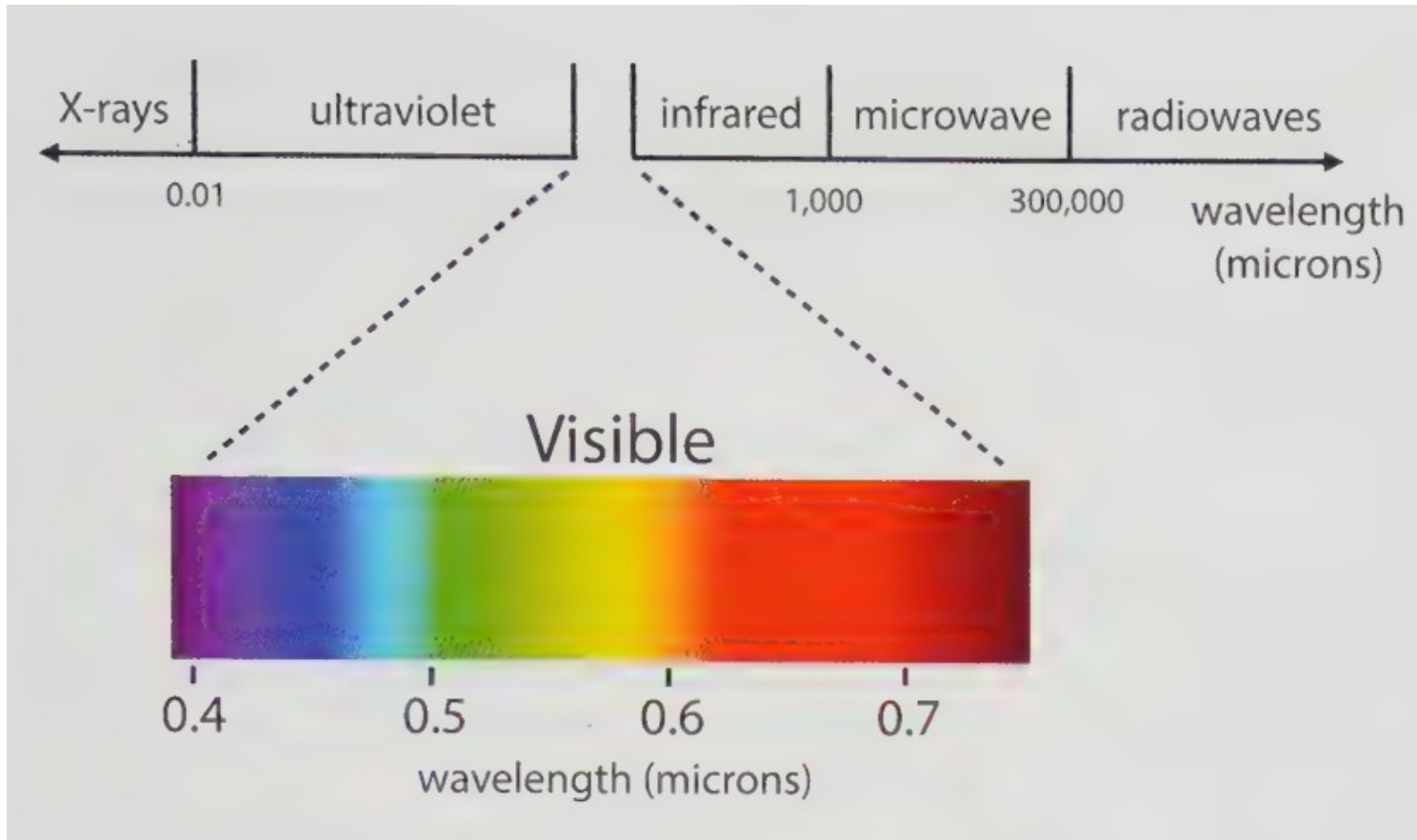


Fig 1. The electromagnetic spectrum

- Visible light = a narrow band within the full spectrum (**Red** → **Violet**)
- Wavelength decreases → frequency increases → energy increases. ($E = \frac{hc}{\lambda}$)

The Visible Spectrum - A Closer Look



Visible Band: 0.4 - 0.8 μm

- Violet/Blue: $\sim 0.4 \mu\text{m}$
- Green/Yellow: $\sim 0.5 - 0.6 \mu\text{m}$
- Orange/Red: $\sim 0.7 - 0.8 \mu\text{m}$

Boundary Points:

UV below 0.4 μm

Infrared above 0.8 μm

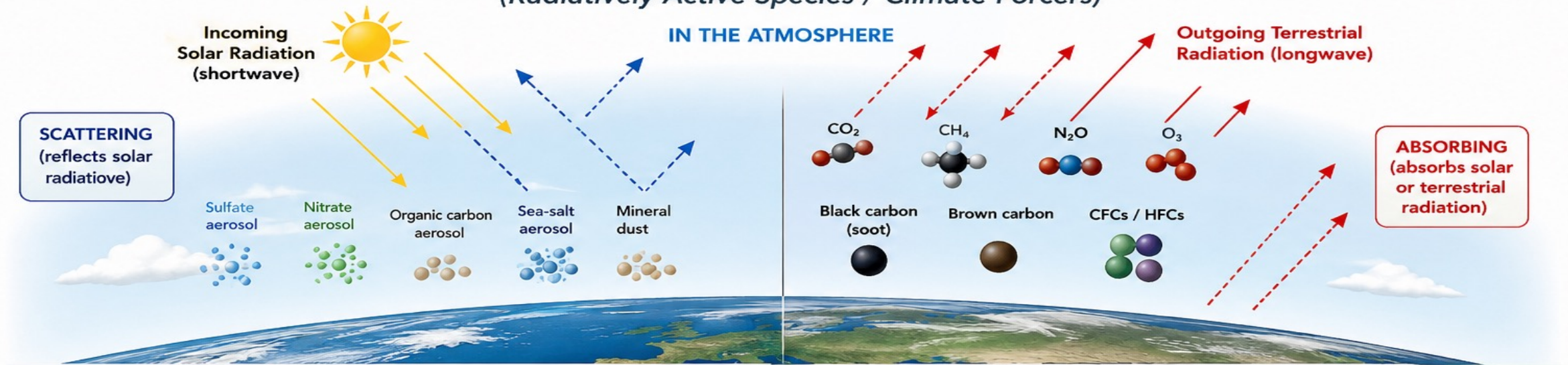
Fig 2. The Visible Spectrum - A Closer Look

Full spectrum axis (wavelength, μm):

X-rays $\rightarrow$ Ultraviolet (0.01 μm) $\rightarrow$ **Visible** $\rightarrow$ Infrared (1,000 μm) $\rightarrow$ Microwave (300,000 μm) $\rightarrow$ Radiowaves

POLLUTANTS ACTIVE IN PERTURBATION OF RADIATION

(Radiatively Active Species / Climate Forcers)



Pollutant / Species	Type	Main Radiative Effect	Typical Climate Effect
● Carbon dioxide (CO ₂)	Gas	Absorbs outgoing longwave (IR) radiation	Warming
● Methane (CH ₄)	Gas	Absorbs longwave; affects O ₃ and stratospheric H ₂ O	Warming
● Nitrous oxide (N ₂ O)	Gas	Absorbs longwave radiation	Warming
● Tropospheric ozone (O ₃)	Gas	Absorbs terrestrial and solar radiation	Warming
● Black carbon (BC)	Aerosol	Strongly absorbs solar radiation	Warming
● Brown carbon (BrC)	Aerosol	Absorbs UV and visible radiation	Warming
● Sulfate aerosol	Aerosol	Strongly scatters sunlight; modifies clouds	Cooling
● Nitrate aerosol	Aerosol	Scatters sunlight	Cooling
● Organic carbon (OC)	Aerosol	Mostly scatters sunlight	Cooling
● Ammonium aerosol	Aerosol	Primarily contributes to scattering aerosol	Cooling
● Mineral dust	Aerosol	Scatters & absorbs shortwave and longwave	Warming or Cooling
● Sea-salt aerosol	Aerosol	Scatters solar radiation; affects clouds	Cooling
● CFCs / HFCs	Gas	Strong absorption of terrestrial IR radiation	Warming

RADIATIVE PRECURSORS (Indirect Effects)

Not strongly radiatively active themselves, but influence radiation by producing or modifying other climate forcers and clouds.

- NO_x → Produces O₃, affects CH₄, nitrate aerosol
- CO → Produces O₃, affects CH₄
- SO₂ → Forms sulfate aerosol
- NH₃ → Forms ammonium aerosol
- NMVOCs → Form O₃ and secondary organic aerosol

- ↑ Warming effect (positive radiative forcing)
- ↓ Cooling effect (negative radiative forcing)

Short-lived climate forcers of particular concern: Black carbon, tropospheric ozone, methane, and some aerosols. Aerosols such as sulfate, nitrate and much of organic aerosol generally exert negative radiative forcing (cooling).

Blackbody Radiation

- Blackbody = idealized object that absorbs **all** incident radiation (no reflection/transmission)
- Emitted radiation (intensity + spectral distribution) depends **only on temperature**, not on material composition

Note: As temperature increases:

- ❖ Total emitted energy **increases**
- ❖ Peak wavelength shifts toward **shorter wavelengths**

 **Sun (≈ 5800 K):**

Emits mainly visible and near-infrared radiation

Peak wavelength $\approx 0.5 \mu\text{m}$

 **Earth (≈ 288 K)**

Emits mainly thermal infrared radiation

Peak wavelength $\approx 10 \mu\text{m}$

❖ Why Room - Temperature Objects Appear “**Black**”:

300 K (room temp) $\rightarrow \lambda_{\text{max}} \approx 9.7 \mu\text{m}$ (far infrared) $\rightarrow$ invisible to humans (visible range: 0.4 - 0.8 μm)



Cont.

❖ Wien's Displacement Law:

Wien's Displacement Law relates the temperature of a blackbody to the wavelength at which it emits the maximum radiation.

$$\lambda_{\max} = \frac{2897}{T} \quad \text{eq. 3}$$

where, $\lambda_{\max}$ in μm & T in Kelvin

➤ Temperature determines the peak wavelength

❖ The Stefan - Boltzmann Equation:

$$\frac{P}{a} = \sigma T^4 \quad \text{eq. 4}$$

where,

$\frac{P}{a}$ = Emitted power per unit area
(W/m^2)

T = Absolute temperature (K)

σ = Stefan - Boltzmann constant
 $= 5.67 \times 10^{-8} \text{ W}/\text{m}^2/\text{K}^4$

➤ A small increase in temperature produces a **large increase in emitted radiation** ($P \propto T^4$)

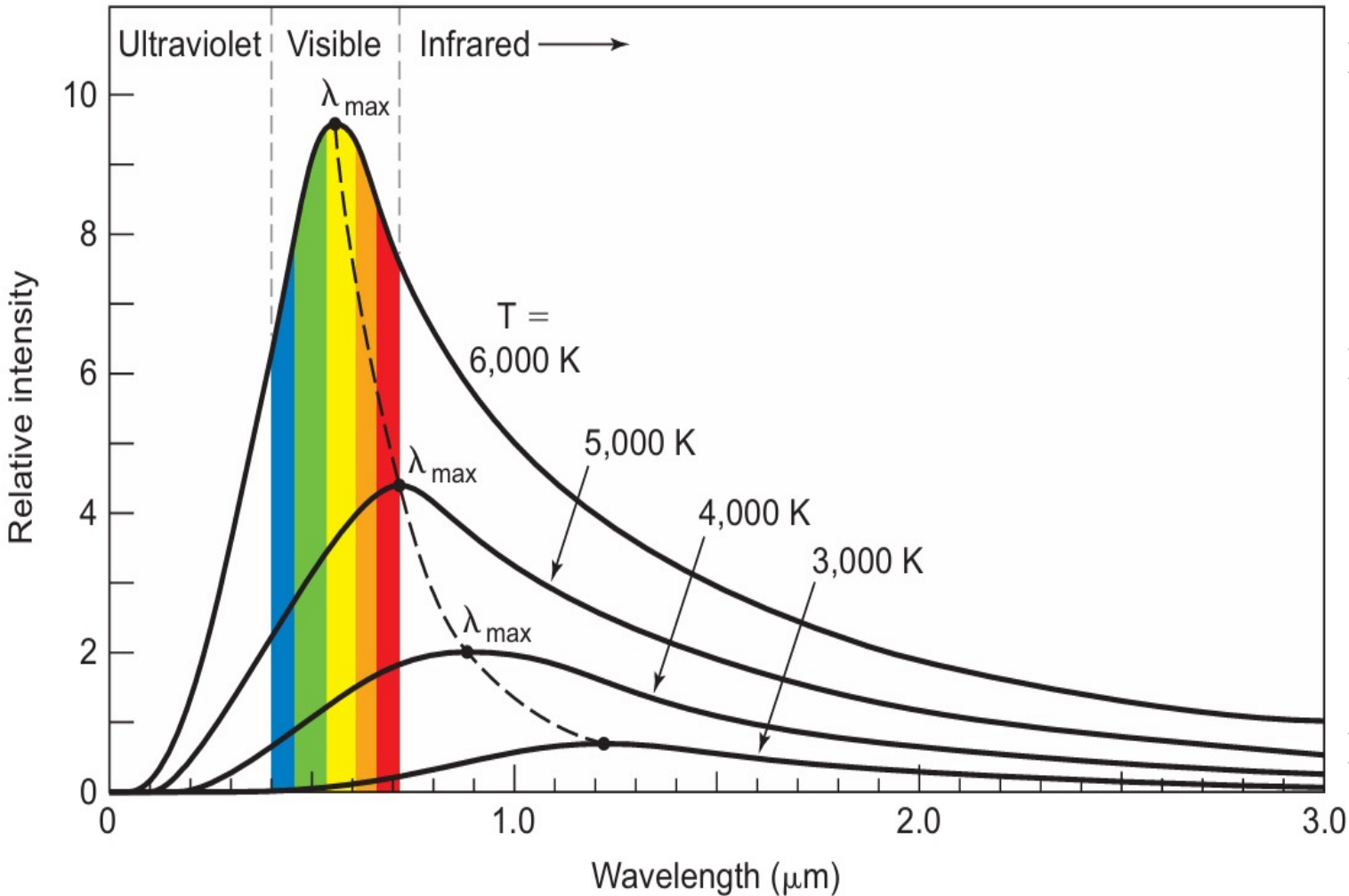


Fig 3. The intensity and spectral distribution of blackbody radiation at various temperatures

Key visual takeaway:

- Only the **6,000 K curve** ($\approx$ Sun's surface temperature) has its peak falling inside the visible band (colored region).
- Cooler blackbodies (3,000-5,000 K) peak in the infrared, with only a small tail extending into visible wavelengths, consistent with objects appearing dim red rather than white-hot.
- This directly explains **why hotter objects glow "whiter"** while cooler one glow "redder"; both the position and height of the curve shift together with temperature.



Blackbody Radiation in Everyday Life

Incandescent Bulb: Temperature Determines What We See

- Electric current heats a tungsten filament to approximately 3000 K, and the hot filament behaves like a blackbody.
- It emits radiation over a broad range of wavelengths.
- Only a small fraction falls within the visible range (0.4 - 0.7 μm).
- Approx. 85% of emitted photons are infrared $\rightarrow$ invisible to the human eye.

Electrical Energy $\rightarrow$ Heat $\rightarrow$ Visible Light (Useful) + Infrared (Mostly wasted for lighting)

How can efficiency be increased?

Increasing filament temperature $\rightarrow$ peak shifts toward shorter wavelengths $\rightarrow$ **greater fraction of radiation becomes visible**

Note: A tungsten filament cannot practically operate near **6000 K** because of material limitations

Practical Solution: **Change the Technology**

- **CFL/LED** bulbs use non-blackbody mechanisms to emit mostly in the visible range
 - Designed to produce useful visible light

Electricity → Electronic Light Emission → Visible Light

- LED and CFL technologies improve lighting efficiency by converting electrical energy into visible light through mechanisms that do not require heating a filament to produce a thermal spectrum

Highlight:

Radiation Physics → Energy Balance → Earth's Climate

- Temperature determines what radiation is emitted, while energy balance determines whether a system warms, cools, or remains in equilibrium.



Energy Balance

First Law of Thermodynamics:

- Energy is conserved
- Photon emission → emitting object's internal energy **decreases** → object **cools**
- Photon absorption → absorbing object's internal energy **increases** → object **warms**

Governing Relation:

The change in internal energy is,

where,

E_{in} = Absorbs radiation

E_{out} = Emits radiation

$$\Delta(\text{Internal Energy}) = E_{in} - E_{out} \quad \text{eq. 5}$$

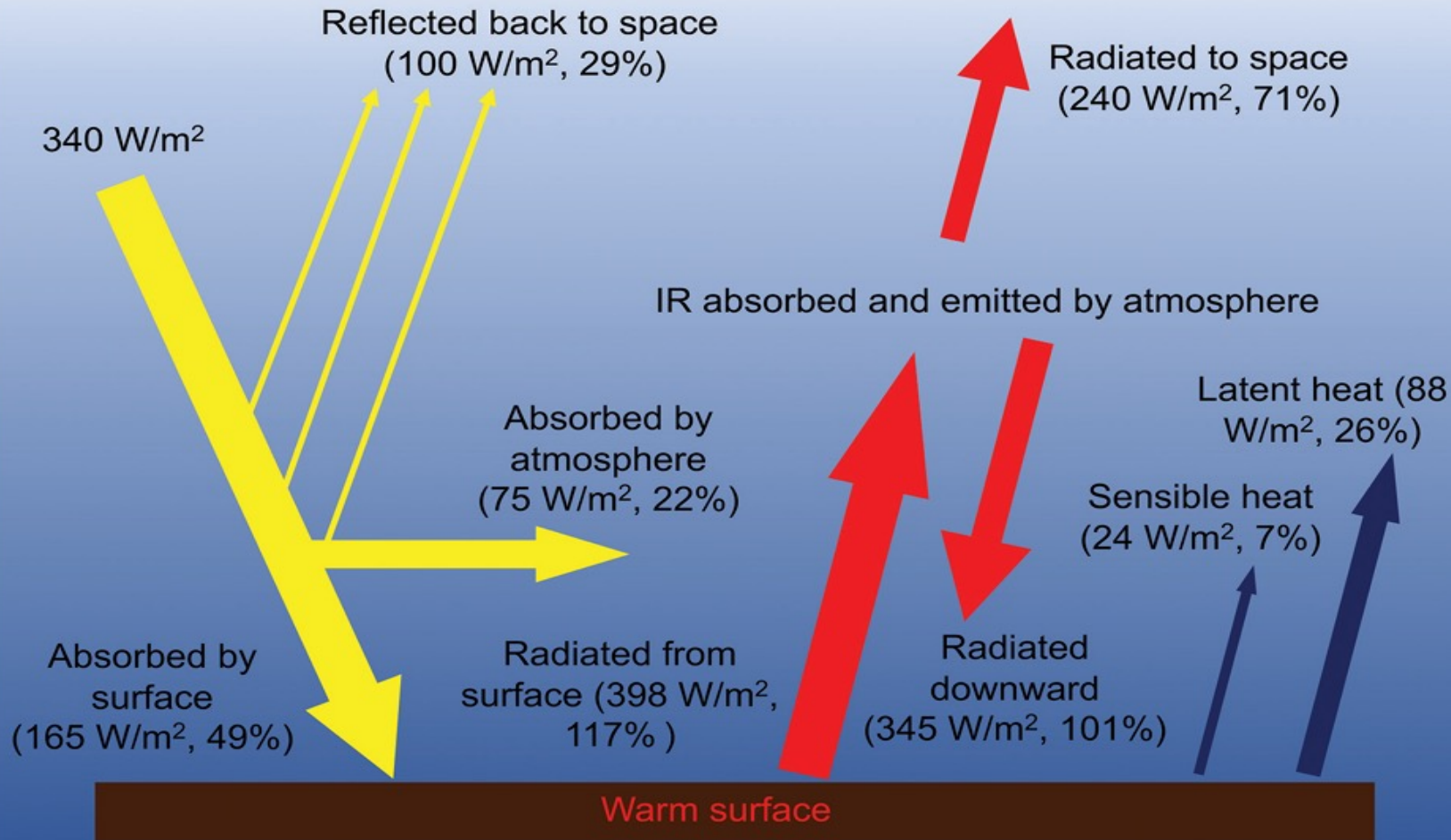
Table 1. Energy Balance Conditions

Condition	Internal Energy	Temperature
$E_{in} > E_{out}$	Increases	Warming
$E_{in} < E_{out}$	Decreases	Cooling
$E_{in} = E_{out}$	Constant	Equilibrium

Cont.



Incoming – outgoing $\approx 0.6 \text{ W/m}^2$



- Earth receives **340 W/m²** of solar radiation on average; roughly half (**165 W/m², 49%**) is absorbed by the surface.
- Surface radiates far more (**398 W/m²**) than it receives directly from the Sun because the atmosphere sends back **345 W/m²** as downward IR radiation.
- At the top of the atmosphere, incoming ($\sim 340 \text{ W/m}^2$) and outgoing (reflected + emitted) are nearly balanced, but not exactly.

Temperature → Radiation Spectrum → Energy Transfer → Energy Balance → Climate

- Earth's climate is fundamentally governed by the balance between incoming solar radiation and outgoing terrestrial infrared radiation
 - Radiation → energy transported by electromagnetic waves/photons
 - Wien's Law: Higher temperature → shorter peak wavelength
 - Stefan - Boltzmann Law: Higher temperature → greater total radiation
 - Hot objects → shorter wavelength + more emitted energy
 - Cool objects → longer wavelength + less emitted energy
- ❖ Radiation controls energy transfer, energy balance controls temperature



Thank You

